

Physics of Complex Systems Homework 1

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Problem 1 (Drunk Driver Statistics (2P)). According to German police statistics, the age of the drunk drivers involved in car accidents with physical injuries are distributed as follows:

Age	18-24	25-34	35-44	45-54	55-64	≥ 65
%	26.1	22.4	18.9	18.5	8.7	5.4

At given age, the percentage of female drivers involved in such accidents is fairly low:

Age	18-24	25-34	35-44	45-54	55-64	≥ 65
%	9.1	11.9	15.2	14.4	12.3	9.2

- (a) What is the probability of being female in such an accident? (1P)
- (b) Use Bayes formula to compute the probability of having such an accident depending on the age given that the driver is female. Where is the maximum? (1P)

Proof. (a) To unpack the mildly unclear language in the description of the tables: Given that someone is in a car accident, the probability that they are in each age group is given in the first table.

Of all the car accidents, the percentage involving female drivers is given in the second table. This is as opposed to the percentage of female drivers involved in such accidents. In other words

$$\begin{aligned} \text{Table 2} &\ni \frac{\text{total number of female drivers in accident}}{\text{total number of drivers in accident}} \\ &\neq \frac{\text{total number of female drivers in accident}}{\text{total number of female drivers}} \end{aligned}$$

We thus calculate, for each age group i (we will say age i , but this will mean an age group as in the table),

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$$\begin{aligned}
P(\text{female, accident}) &= \sum_i P(\text{female, accident} | \text{age } i) P(\text{age } i) \\
&= 10^{-4} (9.1 \cdot 26.1 + 11.9 \cdot 22.4 + 15.2 \cdot 18.9 \\
&\quad + 14.4 \cdot 18.5 + 12.3 \cdot 8.7 + 9.2 \cdot 5.4) \\
&= 0.1214
\end{aligned}$$

(b) We state Bayes's theorem

$$P(\text{accident} | \text{female}) = \frac{P(\text{female} | \text{accident}) P(\text{accident})}{P(\text{female})}$$

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Problem 2 (Poisson Distribution (6P)). The Poisson distribution

$$P_\lambda(k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

can be understood as the limit of the binomial distribution in the case of “rare events”.

(a) Let $p = \lambda/N$ and take $N \rightarrow \infty$ while keeping λ and k constant. Show that in this limit we can approximate $(1 - p)^{N-k} \approx e^{-\lambda}$. (1P)

(b) Show similarly that

$$\binom{N}{k} \approx \frac{N^k}{k!}.$$

(1P)

(c) Use (a) and (b) to show that in this limit, the binomial distribution tends to the Poisson distribution. (1P)

(d) Check that the Poisson distribution is properly normalized. (1P)

(e) Compute the moment- and cumulant-generating functions (MGF and CGF). (1P)

(f) Determine all cumulants. (1P)

Proof. (a) We recall the definition

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n.$$

Then we have

$$\begin{aligned}
 \lim_{N \rightarrow \infty} \left(1 - \frac{\lambda}{N}\right)^{N-k} &= \lim_{N \rightarrow \infty} \left(1 - \frac{\lambda}{N}\right)^N \left(1 - \frac{\lambda}{N}\right)^{-k} \\
 &= \left[\lim_{N \rightarrow \infty} \left(1 - \frac{\lambda}{N}\right)^N \right] \left[\lim_{N \rightarrow \infty} \left(1 - \frac{\lambda}{N}\right)^{-k} \right] \\
 &= \left[\lim_{N \rightarrow \infty} \left(1 + \frac{1}{N}\right)^{-\lambda N} \right] \\
 &= \lim_{N \rightarrow \infty} \left[\left(1 + \frac{1}{N}\right)^N \right]^{-\lambda} \\
 &= \left[\lim_{N \rightarrow \infty} \left(1 + \frac{1}{N}\right)^N \right]^{-\lambda} \\
 &= e^{-\lambda}
 \end{aligned}$$

(b) We have as an exact expression

$$\binom{N}{k} = \frac{N!}{k!(N-k)!}.$$

Since k is held constant, we only need to show that

$$\frac{N!}{(N-k)!} \xrightarrow{N \rightarrow \infty} N^k$$

or alternatively

$$\frac{N!}{N^k(N-k)!} \xrightarrow{N \rightarrow \infty} 1.$$

We do this by

$$\begin{aligned}
 \frac{N!}{N^k(N-k)!} &= \frac{1}{N^k} N(N-1) \dots (N-k+1) \\
 &= 1 \left(1 - \frac{1}{N}\right) \left(1 - \frac{2}{N}\right) \dots \left(1 - \frac{k-1}{N}\right) \\
 &\xrightarrow{N \rightarrow \infty} 1
 \end{aligned}$$

(c) The binomial distribution is given by the PDF

$$P_{N,p}(k) = \binom{N}{k} p^k q^{N-k}.$$

We take $p = \lambda/N$ and approximate

$$\begin{aligned}
 P_{N,p}(k) &\xrightarrow{N \rightarrow \infty} \frac{N^k}{k!} \left(\frac{\lambda}{N}\right)^k e^{-\lambda} \\
 &= \frac{\lambda^k}{k!} e^{-\lambda}
 \end{aligned}$$

(d) The total probability must sum up to 1:

$$\begin{aligned}\sum_{k=0}^{\infty} P_{\lambda}(k) &= \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} e^{-\lambda} \\ &= e^{-\lambda} e^{\lambda} = 1\end{aligned}$$

using the Taylor series for $\exp(x)$.

(e) The MGF is given by

$$\begin{aligned}M_X(t) &= \langle \exp(tX) \rangle \\ &= \sum_{x=0}^{\infty} e^{tx} P(X=x) \\ &= \sum_{x=0}^{\infty} e^{tx} \frac{\lambda^x}{x!} e^{-\lambda} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda \exp(t))^x}{x!} \\ &= e^{-\lambda} e^{\lambda \exp(t)} \\ &= e^{\lambda[\exp(t)-1]}\end{aligned}$$

The CGF is the logarithm of the MGF:

$$K_X(t) = \ln M_X(t) = \lambda[\exp(t) - 1]$$

(f) The cumulants are the coefficients of the Taylor series in K_X :

$$K_X(t) = \sum_{n=1}^{\infty} \frac{\kappa_n}{n!} t^n.$$

Thus, we expand

$$\begin{aligned}K_X(t) &= \lambda \left[\left(\sum_{k=0}^{\infty} \frac{t^k}{k!} \right) - 1 \right] \\ &= \lambda \sum_{k=1}^{\infty} \frac{t^k}{k!}\end{aligned}$$

from which we can read off all cumulants other than κ_0 as being equal to λ . $\square$

Problem 3 (Reconstruct Probabilities from Given Moments (4P)). Consider a system with three configurations $\Omega = \{a, b, c\}$ together with an associated map $X : \Omega \rightarrow \mathbb{R}$ defined by

$$X_a = 1, \quad X_b = 3, \quad X_c = 5.$$

- (a) How many moments do you have to know in order to be able to reconstruct the probability distribution $\{P_a, P_b, P_c\}$ and why?
- (b) Suppose that M_1 and M_2 are given (with the map X defined above). Find the solution for the probabilities P_a, P_b , and P_c .
- (c) Not all possible values for M_1 and M_2 lead to consistent results. Give a counterexample.
- (d) Let $M_1 = \frac{7}{2}$ and $M_2 = 15$. Compute the probabilities P_a, P_b, P_c and find a general formula for the higher moments M_n with $n = 3, 4, \dots, \infty$.

Proof. (a) Since we are in a discrete probability space (in particular, with the discrete measure), the probability measure is defined by the measure of the three point sets $P(\{a\})$, $P(\{b\})$ or $P(\{c\})$.

Thus, if we are given the first two moments, we can construct two equations; the normalisation of probability yields a third. We can solve these three equations to find the 3 probabilities.

- (b) We explicitly write the equations

$$\begin{aligned} 1 &= P_a + P_b + P_c \\ M_1 &= X_a P_a + X_b P_b + X_c P_c \\ M_2 &= X_a^2 P_a + X_b^2 P_b + X_c^2 P_c \end{aligned}$$

This is a linear system of equations

$$\begin{pmatrix} 1 & 1 & 1 \\ X_a & X_b & X_c \\ X_a^2 & X_b^2 & X_c^2 \end{pmatrix} \begin{pmatrix} P_a \\ P_b \\ P_c \end{pmatrix} = \begin{pmatrix} 1 \\ M_1 \\ M_2 \end{pmatrix}.$$

On a side note, this simple formulation provides another easy proof for the moment problem on a discrete distribution and the conditions therefore: The matrix is an example of a Vandermonde matrix, which is invertible iff all X_i s are distinct (obviously, if any two were not distinct, we would have a repeated column, forcing the determinant to vanish).

We can then invert this matrix to find

$$\begin{aligned}
 \begin{pmatrix} 1 & 1 & 1 \\ X_a & X_b & X_c \\ X_a^2 & X_b^2 & X_c^2 \end{pmatrix}^{-1} &= \left[X_a^2(-X_b) + X_a X_b^2 + X_a^2 X_c - X_a X_c^2 + X_b X_c^2 - X_b^2 X_c \right]^{-1} \\
 &\cdot \begin{pmatrix} X_b X_c^2 - X_b^2 X_c & X_b^2 - X_c^2 & X_c - X_b \\ X_a^2 X_c - X_a X_c^2 & X_c^2 - X_a^2 & X_a - X_c \\ X_a X_b^2 - X_a^2 X_b & X_a^2 - X_b^2 & X_b - X_a \end{pmatrix} \\
 &= \begin{pmatrix} \frac{15}{8} & -1 & \frac{1}{8} \\ -\frac{5}{4} & \frac{3}{2} & -\frac{1}{4} \\ \frac{3}{8} & -\frac{1}{2} & \frac{1}{8} \end{pmatrix}
 \end{aligned}$$

which, when applied to the moment vector, yields

$$\begin{pmatrix} P_a \\ P_b \\ P_c \end{pmatrix} = \begin{pmatrix} -M_1 + \frac{M_2}{8} + \frac{15}{8} \\ \frac{3M_1}{2} - \frac{M_2}{4} - \frac{5}{4} \\ -\frac{M_1}{2} + \frac{M_2}{8} + \frac{3}{8} \end{pmatrix}.$$

- (c) Clearly, $M_1 = M_2 = 0$ would make P_b negative, which is not allowed (this would be clear because all values of X are strictly positive, thus the expectation value cannot be zero).
- (d) Substituting in the values yields $P_a = 1/4$, $P_b = 1/4$, $P_c = 1/2$.

The general moment is given by

$$\begin{aligned}
 M_n &= X_a^n P_a + X_b^n P_b + X_c^n P_c \\
 &= 1^n \left(\frac{1}{4} \right) + 3^n \left(\frac{1}{4} \right) + 5^n \left(\frac{1}{2} \right) \\
 &= \frac{1}{4} + \frac{3^n}{4} + \frac{5^n}{2}.
 \end{aligned}$$

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